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In [2]: import pandas as pd
import numpy as np
from google.colab import drive
drive.mount('/content/drive')
df = pd.read_csv('/content/drive/My Drive/ADS/supermarket_sales - Sheet1.csv')
```

Mounted at /content/drive

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In [ ]: df.head()
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Out[]:

	Invoice ID	Branch	City	Customer type	Gender	Product line	Unit price	Quantit
0	750-67-8428	A	Yangon	Member	Female	Health and beauty	74.69	
1	226-31-3081	C	Naypyitaw	Normal	Female	Electronic accessories	15.28	
2	631-41-3108	A	Yangon	Normal	Male	Home and lifestyle	46.33	
3	123-19-1176	A	Yangon	Member	Male	Health and beauty	58.22	
4	373-73-7910	A	Yangon	Normal	Male	Sports and travel	86.31	

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In [51]: from re import S
import math as m
# we will do a hypothesis on total as total amount can be above a certain amount
# z test when pop size > 30 and t test will be when pop size < 30

# Null hypo will be that total amt <= 250
# alternate hypo will total amt > 250
df1 = df.sample(n=28)
df2=df.sample(n=100)

#sample size
nt = len(df1)
nz= len(df2)

#sample mean
x_bar_t= df1["Total"].mean()
x_bar_z= df2["Total"].mean()

# Population size
n=len(df)

# pop mean
meu = df["Total"].mean()

# STANDARD DEVIATION
s_t = df1["Total"].std()
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s_z = df2["Total"].std()
sigma = df["Total"].std()

# formula for Z test
print("for Z test")
Z_SCORE = (x_bar_z-meu)/(sigma/m.sqrt(nz))
# Z_SCORE = round(Z_SCORE,1)
print(Z_SCORE)
alpha = 0.05

critical_val = 1.65 #VALUE FOR RIGHT TAIL AND SIGNIFICANCE
# if z score is greater than critical val then reject null hypothesis
if Z_SCORE > critical_val:
    print("We reject Null Hypothesis")
else :
    print("We Do NOT REJECT Null Hypothesis")

# formula for T test

print("for T test")
T_SCORE = (x_bar_t-meu)/(s_t/m.sqrt(nt))
# T_SCORE = round(T_SCORE,1)
print(T_SCORE)
alpha = 0.05

critical_val = 1.703 # N-1 =27 AND 0.05 KA INTERSECTION
# if z score is greater than critical val then reject null hypothesis
if T_SCORE > critical_val:
    print("We reject Null Hypothesis")
else :
    print("We Do NOT REJECT Null Hypothesis")

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322.96674900000005
0.322966749
322.966749
for Z test
0.10609252072429469
We Do NOT REJECT Null Hypothesis
for T test
0.2309905418834541
We Do NOT REJECT Null Hypothesis

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In [ ]: # FOR 2 SMPLE TEST WE CAN CREATE DATA FRAME FOR MEN AND WOMEN ON GENDER FOR
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In [61]: # Two sample Independent T Test
men_arr=[]
women_arr=[]
import statistics ,math
# for t test
for i in range(29):
    if df["Gender"][i]=="Female":
        women_arr.append(df["Quantity"][i])

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else:
    men_arr.append(df["Quantity"][i])

# means
men_mean= statistics.fmean(men_arr)
women_mean= statistics.fmean(women_arr)

# print(men_mean)
# print(women_mean)

#std dev if xbar is not given then std dev for both men and women comes out to
men_std = statistics.stdev(men_arr,xbar=men_mean)
women_std = statistics.stdev(women_arr,xbar=women_mean)
# print(men_std)
# print(women_std)

# lengths
men_len = len(men_arr)
women_len =len(women_arr)

# formula

lower = math.sqrt((1/men_len)+(1/women_len))
s = math.sqrt((((men_len-1)*men_std*men_std)+((women_len-1)*women_std*women_std))
print(s)

T_SCORE = abs(men_mean-women_mean)/(s*lower)
print(T_SCORE)
# tscore cross verified by hand calc

degree_freedom = men_len+women_len-2
# we check degree_freedom=27 and alpha = 0.05
tcritical = 1.703

if T_SCORE > tcritical:
    print("We reject Null Hypothesis")
else :
    print("We Do NOT REJECT Null Hypothesis")

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14 15 2.5559669467673154 2.6421491973301547
2.6010105213017494
0.4089096286857567
We Do NOT REJECT Null Hypothesis

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In [62]: # Two sample Independent Z Test
# null hypo = me1>me2
# alternate hypo =me1 > me2 so right tailed test at alpha = 0.05

men_arr=[]
women_arr=[]
import statistics ,math
# for t test
for i in range(100):
    if df["Gender"][i]=="Female":

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    women_arr.append(df["Quantity"][i])
else:
    men_arr.append(df["Quantity"][i])

# means
men_mean= statistics.fmean(men_arr)
women_mean= statistics.fmean(women_arr)

# print(men_mean)
# print(women_mean)

#std dev if xbar is not given then std dev for both men and women comes out to
men_std = statistics.stdev(men_arr,xbar=men_mean)
women_std = statistics.stdev(women_arr,xbar=women_mean)
# print(men_std)
# print(women_std)

# lengths
men_len = len(men_arr)
women_len =len(women_arr)

# formula  $x_1-x_2/s$ 

lower = math.sqrt((men_std*men_std/men_len)+(women_std*women_std/women_len))

Z_SCORE = abs(men_mean-women_mean)/lower
print(Z_SCORE)

Zcritical = 1.645

if Z_SCORE > Zcritical:
    print("We reject Null Hypothesis")
else :
    print("We Do NOT REJECT Null Hypothesis")

```

0.056438999111823006

We Do NOT REJECT Null Hypothesis